

ABSTRACT

The current research focuses on the geochemistry of sedimentary rocks found in Cox's Bazar District, located in the southeastern part of Bangladesh. The main objectives of the study are to determine the origin, tectonic setting, characteristics of the source rocks, degree of chemical weathering, and hydraulic sorting processes of the deposits. The analyses showed that the concentration of SiO_2 ranged from 65.5% - 75.2%, Al_2O_3 15.6%–24.8%, Fe_2O_3 0.368–6.01%, K_2O 0.334–3.28%, MgO 0.694–1.91%, CaO 0.0479–1.72%, Na_2O 0.12–2.132%, TiO_2 0.0786–0.971%, P_2O_5 0.0819–0.488% and MnO 0.0027–1.28%. A–CN–K diagrams show an intense weathering trend and a backward trend. The major element provenance discriminant diagram refers to the mature polycyclic quartzes provenance and originated dominantly from felsic to intermediate igneous rocks. The trend of the $\text{SiO}_2/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ shows the hydraulic sorting effect and sediments originated primarily from a recycled sedimentary provenance. The CIA and ICV of the analysed sample ranged from 75–97 and 0.1–0.7 with an average 85 and 0.42, respectively. $\text{K}_2\text{O}\text{--Na}_2\text{O}$ ratios show a moderate to high maturity of the sediments and are derived from both weak and intensively weathered source rocks. Discriminate diagrams related to tectonic provenance refer to the deposit of sediment from active to passive continental margin. The sediments of the study area are recycled sedimentary provenance and derived from the Himalaya after upliftment, erosion, and deposition in the basin.

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